

KEFAN ZHANG

Senior Software Engineer

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SUMMARY

Senior Software Engineer specializing in large-scale ad tech and data pipelines with 10+ years of experience at Netflix and Meta. Proven success in building real-time ad-serving systems, analytics dashboards, and optimization frameworks using React, GraphQL, Java, and Kafka. Passionate about leveraging data engineering and distributed systems to improve monetization, performance, and user experience.

SKILLS

React	React Native	Next.js	Python	TypeScript	JavaScript	Relay	D3.js	Node.js	Java	C#	C++
Hack(PHP)	GraphQL	REST APIs	gRPC	MySQL	AWS	Microsoft Azure	Docker	GitHub Actions	Kubernetes	Kafka	
Jest	Cypress	JUnit	PyTest								

EXPERIENCE

05/2023 - Present

Senior Software Engineer



Netflix

Netflix Ad-Supported Plan

- Engineered dynamic ad-insertion pipelines in Netflix's playback stack using React.js, TypeScript, Node.js, and GraphQL, ensuring seamless ad delivery across devices.
- Integrated real-time targeting models into the ad-supported tier backend using Java, gRPC, and Apache Kafka, scaling to hundreds of thousands of ad requests per second with high availability.
- Developed ad-preview and reporting dashboards in Next.js + D3.js for advertisers, driving adoption of self-service ad campaign creation.
- Built adaptive ad-tracking components in React Query with incremental rendering, significantly improving load times for advertiser dashboards.
- Deployed services on AWS EKS and Spinnaker, with CI/CD pipelines in GitHub Actions + Jenkins, accelerating release cycles.
- Designed automated integration tests using JUnit, Cypress, and Jest, strengthening reliability of ad-serving components.
- Partnered with data science and product teams to launch ad frequency capping, improving user satisfaction in the ad-supported plan.
- Contributed to cross-platform DRM compatibility in playback stack, ensuring compliance with industry standards.
- Mentored junior engineers, introduced Architecture Decision Records (ADRs) in Confluence, and streamlined cross-team RFC reviews.

01/2020 - 05/2023

Software Engineer



Meta

Instagram Story Ads Optimization

- Developed Instagram Story Ads optimization logic using React Native, Python (PyTorch models), and GraphQL APIs, increasing ad engagement.
- Designed advertiser-facing campaign budget dashboard in React.js + Relay, enabling near real-time spend adjustments.
- Extended Meta's ads ranking service in C++ and Hack (PHP), reducing inference latency for ML models.
- Constructed Oculus and Portal e-commerce features (promo code generator, catalog system) with Node.js + MySQL, powering global product sales.
- Automated A/B testing framework integration with Jest, Selenium, and Meta's internal test harness, reducing experiment rollout risks.
- Migrated ad optimization pipelines to Meta's internal container cloud (Tupperware), improving infrastructure efficiency.
- Coordinated across product, infra, and ads integrity teams to launch sensitive-content filters in Story Ads, improving user trust.
- Partnered with Oculus commerce PMs to deliver subscription-based VR storefront, growing app ecosystem monetization.

08/2018 - 01/2020

Software Engineer



Microsoft

Office 365 Real-Time Co-Authoring

- Implemented collaborative editing synchronization in C# and TypeScript, leveraging SignalR and Azure Service Bus to reduce conflicts.
- Implemented a frontend diff-merge UI in React.js to visualize concurrent document edits and simplify conflict resolution.
- Enhanced autosave functionality with Azure Blob Storage, ensuring document persistence across large user bases.
- Scaled backend services on Azure Kubernetes Service with NGINX Ingress, supporting millions of concurrent sessions.
- Released cross-platform collaboration analytics dashboards, enabling adoption tracking across enterprise accounts.

EDUCATION

2014 - 2018

- Bachelor's Degree in Computer Science
University of Virginia